

# 14B. Leases, Pensions and Income Taxes (Ch 10)



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## 14B.0 Overview

- *Balance sheet quality* = how accurately reported assets and liabilities reflect a firm's true economic position. Strong-balance-sheet pressure can lead firms to *window dress*, often by keeping debt off the books.
- The chapter covers **leases, pensions, and income taxes**

| <b>EXHIBIT 10.1 Deere &amp; Company Balance Sheets (excerpts)</b>           |                  |                  |
|-----------------------------------------------------------------------------|------------------|------------------|
| <b>DEERE &amp; COMPANY</b>                                                  |                  |                  |
| <b>Consolidated Balance Sheet (excerpts)</b>                                |                  |                  |
| <b>As of October 27, 2024 and October 29, 2023 (in millions of dollars)</b> | <b>2024</b>      | <b>2023</b>      |
| <b>ASSETS</b>                                                               |                  |                  |
| Cash and cash equivalents . . . . .                                         | \$ 7,324         | \$ 7,458         |
| Marketable securities . . . . .                                             | 1,154            | 946              |
| Trade accounts and notes receivable—net . . . . .                           | 5,326            | 7,739            |
| Financing receivables—net . . . . .                                         | 44,309           | 43,673           |
| Financing receivables securitized—net . . . . .                             | 8,723            | 7,335            |
| Other receivables . . . . .                                                 | 2,545            | 2,623            |
| Equipment on operating leases—net . . . . .                                 | 7,451            | 6,917            |
| Inventories . . . . .                                                       | 7,093            | 8,160            |
| Property and equipment—net . . . . .                                        | 7,580            | 6,879            |
| Goodwill . . . . .                                                          | 3,959            | 3,900            |
| Other intangible assets—net . . . . .                                       | 999              | 1,133            |
| Retirement benefits . . . . .                                               | 2,921            | 3,007            |
| Deferred income taxes . . . . .                                             | 2,086            | 1,814            |
| Other assets . . . . .                                                      | 2,906            | 2,503            |
| Assets held for sale . . . . .                                              | 2,944            | —                |
| <b>Total Assets</b> . . . . .                                               | <b>\$107,320</b> | <b>\$104,087</b> |
| <b>LIABILITIES AND STOCKHOLDERS' EQUITY</b>                                 |                  |                  |
| <b>LIABILITIES</b>                                                          |                  |                  |
| Short-term borrowings . . . . .                                             | 13,533           | 17,939           |
| Short-term securitization borrowings . . . . .                              | 8,431            | 6,995            |
| Accounts payable and accrued expenses . . . . .                             | 14,543           | 16,130           |
| Deferred income taxes . . . . .                                             | 478              | 520              |
| Long-term borrowings . . . . .                                              | 43,229           | 38,477           |
| Retirement benefits and other liabilities . . . . .                         | 2,354            | 2,140            |
| Liabilities held for sale . . . . .                                         | 1,827            | —                |
| <b>Total liabilities</b> . . . . .                                          | <b>84,395</b>    | <b>82,201</b>    |
| Commitments and contingencies (Note 20)                                     |                  |                  |
| Redeemable noncontrolling interest (Note 3) . . . . .                       | 82               | 97               |
| <b>STOCKHOLDERS' EQUITY</b>                                                 |                  |                  |
| Total stockholders' equity . . . . .                                        | 22,843           | 21,789           |
| <b>Total Liabilities and Stockholders' Equity</b> . . . . .                 | <b>\$107,320</b> | <b>\$104,087</b> |

## Part 1: Leases

### 14B.1 Lease Contracts

- A **lease** is a contractual agreement between a lessor and a lessee that gives the **lessee** the right to use specific property, owned by the **lessor**, in exchange for periodic payments for a specified period of time.
- Many businesses **lease** substantial portions of the property and equipment they use in their business organization as **an alternative to ownership**.
- Leases are popular!

| Company        | Description                                                                                                                                                           |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Apple          | "The Company leases various equipment and facilities, including retail space, under noncancelable operating lease arrangements."                                      |
| ExxonMobil     | "Minimum commitments for operating leases, shown on an undiscounted basis, cover drilling equipment, tankers, service stations, and other properties."                |
| JPMorgan Chase | "JPMorgan Chase and its subsidiaries were obligated under a number of noncancelable operating leases for premises and equipment used primarily for banking purposes." |

Source: Company 10-K filings

- Benefits of Leases to **Lessees**
  - Leasing reduces the upfront cash need to use an asset.
  - Lease payments often are lower than installment payments (to buy an asset)
    - Because the lease period is shorter.
  - Leasing might offer protection against the obsolescence.
  - Leasing may be **less costly financing**
    - **Tax benefits** the lessor received may be passed to the lessee.
- Benefits of Leases to **Lessors**
  - Leasing stimulates sales of a lessor's product
  - Leasing provides **tax benefits** to various parties in the lease, which enhances the return to all the parties involved
    - E.g., Many start-up companies do not have enough taxable income to fully benefit from tax savings from depreciation expense. Through a lease arrangement, the lessor can take depreciation expenses, and this tax savings can be passed to the lessee.
  - Leasing provides a high residual value to the lessor upon the return of the property.

## 14B.2 Lease Classification & Accounting Treatment

- The issue of how to report leases is the classic case of **substance versus form**.
  - Accounting for leases depends on whether
    - 1) the lease arrangement **resembles the purchase/sale of an asset with debt financing** (**finance lease**) or whether
    - 2) it is a **rental agreement** (**operating lease**).

|                  | Finance Lease                                                                                                                             | Operating Lease                                                                               |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Balance sheet    | Right-of-Use (ROU) asset + lease liability                                                                                                | ROU asset + lease liability                                                                   |
| Income statement | Two expenses: <ul style="list-style-type: none"> <li>• amortization (straight-line) +</li> <li>• interest (effective interest)</li> </ul> | One expense: <ul style="list-style-type: none"> <li>• lease expense, straight-line</li> </ul> |
| Expense pattern  | Front-loaded (higher early)                                                                                                               | Level (same each year)                                                                        |
| Cash flow        | <ul style="list-style-type: none"> <li>• Interest = operating</li> <li>• principal = financing</li> </ul>                                 | Entire payment = operating                                                                    |

### Illustration — Lease Classification and its Accounting

Gillette Electronics leases a piece of equipment under the following terms:

- Lease term: **5 years**
- Annual lease payment: **\$10,000**, payable at each year-end
- Discount rate: **7%**
- Present Value of lease payments: \$41,002

| Excel Solution                |             |
|-------------------------------|-------------|
| <i>i</i>                      | 7%          |
| <i>n</i>                      | 5           |
| PMT                           | \$ (10,000) |
| FV                            | \$ -        |
| PV                            | \$ 41,002   |
| PV(rate,nper,pmt,[fv],[type]) |             |

### Case 1: Equipment Life = 5.5 Years

- Assume the leased equipment has an estimated economic life of **5.5 years**.

### Classification Analysis

- One of the finance lease criteria is whether: The lease term covers a **major part of the remaining economic life** of the underlying asset. In this case:

$$\frac{\text{Lease Term}}{\text{Economic Life}} = \frac{5}{5.5} = 90.9\%$$

- Because the lease term covers almost all of the asset's economic life, the lessee effectively obtains control of the asset for most of its useful life.
- Therefore, the lease is classified as a **finance lease** because it functions economically like a purchase financed through installment payments.

**Case 2: Operating Lease**

- Assume the equipment life is 15 years
- Since the 5-year lease term covers only a small portion of the asset's 15-year life (about 33%), the lease is treated as an **operating lease**. Economically, it is **similar to renting the asset**.

| Gillette Electronics (Lessee)<br>Lease Amortization Schedule |                      |                             |                              |                 |
|--------------------------------------------------------------|----------------------|-----------------------------|------------------------------|-----------------|
| Date                                                         | Annual Lease Payment | Interest on Lease Liability | Reduction of Lease Liability | Lease Liability |
| 1/1/2026                                                     |                      |                             |                              | <b>\$41,002</b> |
| 12/31/2026                                                   | 10,000               | 2,870                       | 7,130                        | 33,872          |
| 12/31/2027                                                   | 10,000               | 2,371                       | 7,629                        | 26,243          |
| 12/31/2028                                                   | 10,000               | 1,837                       | 8,163                        | 18,080          |
| 12/31/2029                                                   | 10,000               | 1,266                       | 8,734                        | 9,346           |
| 12/31/2030                                                   | 10,000               | 654                         | 9,346                        | 0               |
|                                                              | <u>\$50,000</u>      | <u>\$8,998</u>              | <u>\$41,002</u>              |                 |

**Finance Lease (Lessee)****Operating Lease (Lessee)***Beginning of the Lease (January 1, 2026)*

|                    |          |
|--------------------|----------|
| Right-of-Use Asset | \$41,002 |
| Lease Liability    | \$41,002 |

|                    |        |
|--------------------|--------|
| Right-of-Use Asset | 41,002 |
| Lease Liability    | 41,002 |

*First lease payment of \$10,000 (December 31, 2026)*

|                  |          |
|------------------|----------|
| Lease Liability  | \$7,130  |
| Interest Expense | \$2,870  |
| Cash             | \$10,000 |

|               |        |
|---------------|--------|
| Lease Expense | 10,000 |
| Cash          | 10,000 |

*To amortize the right-of use asset (December 31, 2026-2030)*

|                    |                      |
|--------------------|----------------------|
| Amort. Expense     | \$8,200 <sup>a</sup> |
| Right-of-Use Asset | \$8,200              |

|                    |         |
|--------------------|---------|
| Lease Liability    | \$7,130 |
| Right-of-Use Asset | \$7,130 |

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<sup>a</sup>. To record amortization of the right-of-use asset (like depreciation expense), applying Sterling's depreciation policy (straight-line). **Right-of-Use Asset Amort**= (\$41,002 ÷ 5 years = \$8,200 /yr)

| Year 1 - 2026                                       | Finance Lease   | Operating Lease |
|-----------------------------------------------------|-----------------|-----------------|
| Interest expense (on lease liability)               | <b>\$2,870</b>  |                 |
| <b>Amortization expense (on right-of-use asses)</b> | <u>8,200</u>    |                 |
| Total Lease Expense                                 | <u>\$11,070</u> | <u>\$10,000</u> |

### What if the company exercised the Loan/Purchase option instead of Lease

#### Finance Lease (Lessee)

#### Loan/Purchase

*Beginning of the Lease (January 1, 2026)*

|                    |            |
|--------------------|------------|
| Right-of-Use Asset | \$41,002 ↑ |
| Lease Liability    | \$41,002 ↑ |

|              |          |
|--------------|----------|
| Cash         | 41,002 ↑ |
| Loan Payable | 41,002 ↑ |
| Equipment    | 41,002 ↑ |
| Cash         | 41,002 ↓ |

*First lease payment of \$10,000 (December 31, 2026)*

|                  |            |
|------------------|------------|
| Lease Liability  | \$7,130 ↓  |
| Interest Expense | \$2,870 ↑  |
| Cash             | \$10,000 ↓ |

|                  |            |
|------------------|------------|
| Lease Liability  | \$7,130 ↓  |
| Interest Expense | \$2,870 ↑  |
| Cash             | \$10,000 ↓ |

*To amortize the right-of use asset (December 31, 2026-2030)*

|                    |                      |
|--------------------|----------------------|
| Amort. Expense     | \$8,200 <sup>a</sup> |
| Right-of-Use Asset | \$8,200              |

|                        |         |
|------------------------|---------|
| Depreciation Expense   | \$8,200 |
| Equipment (Amort. Exp) | \$8,200 |

### Finance vs. Operating — Same Total, Different Timing

| Year  | Finance: Interest | Finance: Amort. | Finance: Total | Operating: Expense |
|-------|-------------------|-----------------|----------------|--------------------|
| 1     | \$2,870           | \$8,200         | \$11,070       | \$10,000           |
| 2     | 2,371             | 8,200           | 10,571         | 10,000             |
| 3     | 1,837             | 8,200           | 10,037         | 10,000             |
| 4     | 1,266             | 8,201           | 9,467          | 10,000             |
| 5     | 654               | 8,201           | 8,855          | 10,000             |
| Total | \$8,998           | \$41,002        | \$50,000       | \$50,000           |

- Over the full term both methods expense the same \$50,000 and fully amortize the asset and liability to zero. The finance lease is front-loaded; the operating lease is level. This is why classification affects reported income timing, leverage, and operating vs. financing cash flow — but never the total cost.

**Concept Quiz**

1. Under ASC 842, a lessee recognizes a \_\_\_\_\_ asset and a \_\_\_\_\_ for essentially all leases at commencement.
2. A finance lease reports two expenses — \_\_\_\_\_ on the ROU asset and \_\_\_\_\_ on the lease liability.
4. An operating lease reports a single \_\_\_\_\_ expense recognized on a \_\_\_\_\_ basis.
5. Over the entire lease term, the total expense under the finance and operating methods is \_\_\_\_\_.

**Discussion Question 1.1 — Lease Classification & Effects**

A company signs a 5-year lease on equipment with annual year-end payments, discounted at its incremental borrowing rate. The asset's economic life is 6 years and ownership does not transfer. This is a finance lease.

**Required:**

- a. Contrast how net income in Year 1 differs under the finance vs. operating method, and why.
- b. Explain why total lease expense over the term is identical under both methods.

## Part 2: Pensions

### 14B.3 Two Types of Pension Plans

- **Defined contribution (DC)**

- Employer/employee contribute to individual accounts (e.g., 401(k)).
- Benefits depend on contributions + investment returns.
- The employer's only expense is its contribution.
- Consequently, accounting is straight forward.

|                 |             |
|-----------------|-------------|
| Pension Expense | \$1,000,000 |
| Cash            | \$1,000,000 |

- **Defined benefit (DB)**

- The employer promises a future benefit (based on years of service and salary).
- The firm bears the investment risk.
- Plans can be over- or underfunded.
- Use has declined sharply, but DB obligations remain large for many firms.
- Accounting is complex.

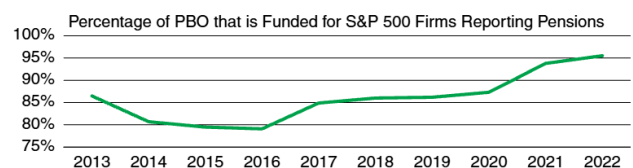
### 14B.4 Accounting for Defined Benefit Plans

- **Pension plan assets** — an investment portfolio (debt and equity securities) funding future retiree payments. It *increases* with investment income and cash contributions, and *decreases* with losses and benefits paid.

- **Projected benefit obligation (PBO)** — the present value of estimated future retiree payments (like the lease liability). Hard to estimate (timing, eligible employees, longevity, final salaries), so firms use actuaries. The PBO *decreases* as benefits are paid.

- **Balance Sheet Reports Funded Status only**

- Funded Status : plan assets minus the PBO:
  - a net asset if **overfunded**, or
  - a net liability if **underfunded** (the more common case).



- **Income Statement – Net Pension Expense**

- **Net pension expense = Service cost + Interest cost – Expected return on plan assets ± Amortization of deferred amounts**

- **Employee Retirement Income Security Act of 1973 (ERISA)** — federal pension law requires plan sponsors (the firm) to set aside money for future benefit payments by funding the plan assets.



## Pension Disclosure From FedEx 10-K

| \$ millions                                              | U.S. Pension Plans |                   | International Pension Plans |                 | Postretirement Healthcare Plans |                  |
|----------------------------------------------------------|--------------------|-------------------|-----------------------------|-----------------|---------------------------------|------------------|
|                                                          | 2023               | 2022              | 2023                        | 2022            | 2023                            | 2022             |
| PBO or APBO at the end of year . . . . .                 | \$26,426           | \$28,702          | \$990                       | \$1,096         | \$ 1,169                        | \$ 1,286         |
| <b>Change in plan assets</b>                             |                    |                   |                             |                 |                                 |                  |
| Fair value of plan assets at the beginning of year . . . | \$25,970           | \$29,785          | \$ 663                      | \$2,133         | \$ —                            | \$ —             |
| Actual return on plan assets . . . . .                   | (707)              | (3,168)           | (83)                        | (4)             | —                               | —                |
| Company contributions . . . . .                          | 826                | 533               | 55                          | 135             | 71                              | 68               |
| Benefits paid . . . . .                                  | (1,263)            | (1,180)           | (35)                        | (36)            | (107)                           | (113)            |
| Settlements . . . . .                                    | —                  | —                 | (20)                        | (1,395)         | —                               | —                |
| Other . . . . .                                          | —                  | —                 | (1)                         | (170)           | 36                              | 45               |
| Fair value of plan assets at the end of year . . . . .   | \$24,826           | \$25,970          | \$ 579                      | \$663           | \$ —                            | \$ —             |
| <b>Funded status of the plans . . . . .</b>              | <b>\$ (1,600)</b>  | <b>\$ (2,732)</b> | <b>\$(411)</b>              | <b>\$ (433)</b> | <b>\$(1,169)</b>                | <b>\$(1,286)</b> |

## Discussion Question 2.1 — Pension Reporting

A company's defined benefit plan has a PBO of \$1,800M and plan assets of \$1,200M. During the year it reports service cost of \$90M, interest cost of \$80M, an expected return of \$95M, and no amortization.

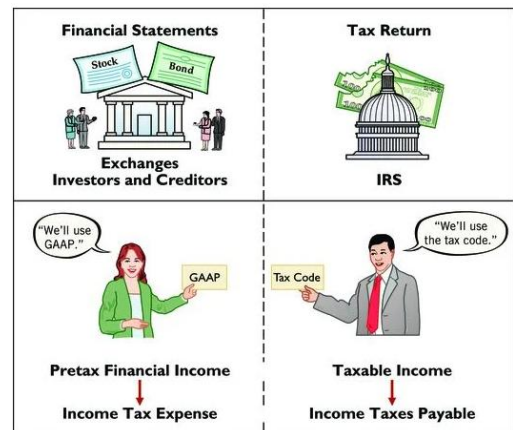
**Required:**

- Compute the funded status and state how it appears on the balance sheet.
- Compute net pension expense for the year.

## Part 3: Accounting for Income Taxes

### 14B.5 Conceptual Underpinnings

- Revenues / expenses / **income** are measured differently for financial reporting purpose (GAAP) and tax purposes (IRS rules).
- Consequently, amounts reported for the following will differ:
  - **income tax expense** (GAAP).
  - **income tax payable** (IRS rules).



### The Basic Idea

- A deferred tax asset is essentially a future tax benefit. It arises when book accounting recognizes a deduction or loss before the tax rules allow you to use it on the tax return.
- **Mental model:** *“As the tax deduction comes later, we need to pay tax for the accounting expenses that are deductible today”*

### Two Sets of Books, Two Clocks

- Every company effectively keeps two income measurements running side by side, prepared under two different rulebooks:
  - **Book income** — accrual, GAAP, estimate-friendly, reported to investors.
  - **Taxable income** — computed on the return, cash-oriented, estimate-hostile, reported to the taxing authority.
- The tax rulebook is generally more skeptical of estimates: it usually wants to see cash actually move before it allows a deduction. GAAP does not wait — it records the expense when the obligation is economically incurred.
- When the two rulebooks disagree about the timing of an item — not about the total amount over its life — someone has to carry that difference on the balance sheet until it unwinds. That carrying account is the deferred tax asset or the deferred tax liability.

### The Connection to Accrual Accounting You Already Know

- This is not a new concept. It is a timing difference of exactly the kind you already handle everywhere else in accrual accounting:
  - **Prepaid expense:** cash is paid now, but the accounting expense is recognized later.
  - **Deferred tax asset:** the accounting expense or loss is recognized now (and tax has been paid), but the tax benefit is realized later.

- In both cases the balance sheet is holding a benefit that has been paid for but not yet used. Deferred taxes simply apply accrual logic to the tax line at the bottom of the income statement.

### Timing Differences vs. Differences That Never Reverse

Only differences that eventually reverse create deferred tax accounts. If a difference never reverses, there is nothing to defer.

|                                         | Temporary difference                                                                           | Permanent difference                                                                |
|-----------------------------------------|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>Does it reverse?</b>                 | Yes — timing only                                                                              | No — never                                                                          |
| <b>Creates a deferred tax account?</b>  | Yes (DTA or DTL)                                                                               | No                                                                                  |
| <b>Effect on the effective tax rate</b> | None over the life of the item                                                                 | Permanently moves the ETR                                                           |
| <b>Examples</b>                         | Accrued bonuses, warranty accrual, depreciation method, unearned revenue, net operating losses | Municipal bond interest, fines and penalties, 50% of meals, life insurance proceeds |

## 14B.6 What a Deferred Tax Asset Is

### A Simple Example

- Suppose a company has \$100 of revenue and records a \$20 expense this year. For financial reporting the expense is recognized now. Assume the tax rules do not allow the \$20 deduction until next year.

|                    | Book accounting | Tax return   |
|--------------------|-----------------|--------------|
| Revenue            | \$100           | \$100        |
| Expense recognized | (\$20)          | \$0          |
| <b>Income</b>      | <b>\$80</b>     | <b>\$100</b> |
| <b>Tax @ 25%</b>   | <b>\$20</b>     | <b>\$25</b>  |

**Result:** the company pays \$25 of current tax even though the tax expense associated with its \$80 of book income is only \$20. That extra \$5 is a tax benefit the company expects to receive in the future.

**Compute it directly:** Deferred tax asset = \$20 future deduction × 25% tax rate = \$5.

### Why Is It an Asset?

Next year, when the tax rules finally allow the \$20 deduction, taxable income will be \$20 lower. At a 25% rate, that deduction saves \$5 of tax. The \$5 deferred tax asset therefore represents an expected reduction in future taxes — a resource the company controls as a result of a past event, which is precisely the definition of an asset.

A deferred tax asset is a receivable from the government. Like any receivable, the real question is whether you will ever collect it.

### Deferred Tax Asset vs. Deferred Tax Liability

| Deferred tax ASSET (DTA)                                              | Deferred tax LIABILITY (DTL)                           |
|-----------------------------------------------------------------------|--------------------------------------------------------|
| <b>Future tax savings</b>                                             | <b>Future tax payments</b>                             |
| Taxable income > book income today                                    | Taxable income < book income today                     |
| You paid the tax early — a deduction is owed to you                   | You deferred the tax — a payment is owed by you        |
| Economically like a prepaid expense or receivable                     | Economically like an accrued liability                 |
| Accrued bonuses, warranty accrual, unearned revenue, NOL carryforward | Accelerated tax depreciation (MACRS vs. straight-line) |

- Both arise for the same underlying reason: financial accounting and tax accounting recognize income and expenses in different periods.

### 14B.7 The Important Catch — Can the Company Actually Use It?

- A deferred tax asset only has economic value if the company expects to have enough taxable income, or another available source of taxable income under the applicable rules, to absorb the deduction. A deduction you can never use is worth nothing.
- If realization is not sufficiently likely under the relevant accounting framework, the reported asset must be reduced:

|                               |                    |
|-------------------------------|--------------------|
| Gross deferred tax asset      | \$10 million       |
| Less: valuation allowance     | (\$7 million)      |
| <b>Net deferred tax asset</b> | <b>\$3 million</b> |

- Under U.S. GAAP, a valuation allowance is recorded when it is **more likely than not** (greater than 50% probability) that some or all of the deferred tax asset will not be realized. If prospects later improve, the company may release part of that

allowance, increasing the net deferred tax asset — and, because the release runs through income tax expense, increasing reported earnings without any change in operations.

### Worked Example – Warranty Accrual

#### Facts

- Book pretax income is \$1,000 in both Year 1 and Year 2.
- In Year 1 the company accrues \$100 of warranty expense for products already sold; no cash is paid.
- In Year 2 the company pays the \$100 of warranty claims; no new accrual is made.
- The enacted tax rate is 21% in both years.

*Under GAAP the expense belongs to Year 1 (matching). Under the tax law the deduction is not allowed until the claim is economically performed and paid in Year 2.*

#### Two-Year Computation

|                                          | Year 1       | Year 2      | Total        | Note                   |
|------------------------------------------|--------------|-------------|--------------|------------------------|
| Book pretax income                       | 1,000        | 1,000       | 2,000        | GAAP                   |
| Add: warranty expense not yet deductible | 100          | —           | 100          | Book-tax adj.          |
| Less: warranty claims paid               | —            | (100)       | (100)        | Reversal               |
| <b>Taxable income</b>                    | <b>1,100</b> | <b>900</b>  | <b>2,000</b> | Per return             |
| Income taxes payable (21%)               | 231          | 189         | 420          | Cash to IRS            |
| Income tax expense (21% × book)          | 210          | 210         | 420          | Income stmt.           |
| <b>Difference → deferred tax asset</b>   | <b>21</b>    | <b>(21)</b> | <b>—</b>     | Created, then reverses |

#### Journal Entries

##### Year 1 — the DTA is created

| Account              | Debit | Credit |
|----------------------|-------|--------|
| Income tax expense   | 210 ↑ |        |
| Deferred tax asset   | 21 ↑  |        |
| Income taxes payable |       | 231 ↑  |

##### Year 2 — the DTA reverses

| Account              | Debit | Credit |
|----------------------|-------|--------|
| Income tax expense   | 210↑  |        |
| Deferred tax asset   |       | 21↓    |
| Income taxes payable |       | 189↑   |

### What the DTA Actually Accomplished

Income tax expense is \$210 in both years even though the cash tax bill bounced from \$231 to \$189. The deferred tax asset is the plug that keeps tax expense matched to book income.

**Conceptual link:** The deferred tax asset does for the bottom of the income statement exactly what prepaid rent does higher up — it is the matching principle applied to the tax line.

## 14B.8 Common Ways Deferred Tax Assets Arise

### Accrued Expenses

A company records employee bonuses in December because the employees earned them in December. Tax law may not allow the deduction until the bonuses are actually paid in February. The book expense is recognized now, the deduction arrives later, and the future deduction creates a deferred tax asset.

### Net Operating Losses

Suppose a start-up generates a \$1 million tax loss. If the tax law allows that loss to offset taxable profits in future years, the loss carries future tax value. At a 25% rate the potential deferred tax asset would be \$250,000 — subject to the applicable carryforward rules and to whether the company expects to be able to use it. At most companies with a loss history, the NOL carryforward is by far the largest single deferred tax asset on the balance sheet.

**Other Common Sources**

| Source                            | Why the tax deduction comes later                                                                                        |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Warranty and bad-debt allowances  | Book accrues an estimate; tax waits for actual payment or write-off                                                      |
| Deferred / unearned revenue       | Tax generally follows cash receipt; book defers to the performance obligation                                            |
| Inventory capitalization (§263A)  | More cost capitalized into tax inventory than into book inventory                                                        |
| Charitable contribution carryover | Deduction limited in the current year and carried forward                                                                |
| Pension and OPEB accruals         | Book expense follows actuarial cost; the deduction follows funding — a direct tie to the pension section of this chapter |

**Discussion Questions****Question 1**

A company accrues \$500,000 of estimated litigation expense in Year 1. The settlement will not be paid until Year 3. The enacted rate is 21%. What deferred tax account arises in Year 1, and for how much?

**Question 2**

A company earns \$80,000 of municipal bond interest, which is never taxable. What deferred tax account arises?

**Question 3**

A start-up has a \$4 million NOL carryforward and a \$840,000 gross deferred tax asset. It has reported losses in each of its first four years and has no signed contracts for next year. How should it report the DTA?

**Question 4**

Why does a company that uses MACRS for tax and straight-line for books end up with a deferred tax liability early in an asset's life, but a shrinking one later?